SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED
CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

Code of Conduct
I Dinesh.S(192521152)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Assessment Tasks
Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud.
   The system must handle millions of concurrent users with minimal buffering.
   Design a cloud architecture using scalability and caching techniques.
   Explain how load balancing and auto-scaling can ensure smooth streaming.
   Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers.
   They face challenges in maintaining consistency and managing deployments.
   Propose a solution using container orchestration and multi-cloud strategies.
   Explain how service discovery and scaling can be handled efficiently.
   Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud.
   The organization needs a robust backup and disaster recovery strategy.
   Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data.
   Some applications must remain on-premises due to compliance requirements.
   Design a hybrid architecture that ensures secure communication between environments.
   Explain how identity management and encryption should be implemented.
   Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application.
   The system must continue operating even if one region goes down.
   Propose a multi-region deployment strategy in the cloud.
   Explain how data replication and failover mechanisms should be configured.
   Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

APPLICATION BASED QUESTION:

1. Auto-Scaling and Load Balancing for Seasonal Traffic

Introduction

An e-commerce company may experience unpredictable traffic during seasonal sales, festivals, and special offers. Cloud computing provides auto-scaling and load balancing to manage sudden increases in users without affecting application performance.

Auto-Scaling

Auto-scaling automatically increases or decreases the number of servers according to the workload. When traffic increases, additional servers are started automatically. When traffic decreases, unnecessary servers are removed. This provides flexibility and also reduces cloud infrastructure costs.

Load Balancing

Load balancing distributes incoming user requests among multiple servers. It prevents a single server from becoming overloaded. If one server fails, the load balancer redirects requests to healthy servers, improving system availability and reliability.

Role of Monitoring

Monitoring tools continuously observe CPU usage, memory, network traffic, response time, and server health. When resource usage becomes high, alerts can be generated and auto-scaling can be triggered. Monitoring also helps administrators identify failures and performance problems.

Real-World Example

During a major festival sale, an e-commerce website may receive millions of customer requests. Auto-scaling adds more servers, while load balancing distributes customers among them. This prevents crashes, reduces downtime, and provides faster page loading and smoother checkout.

Benefits

- Handles sudden traffic increases.
- Reduces downtime.
- Improves application performance.
- Provides better user experience.
- Reduces unnecessary infrastructure costs.

2. Secure Cloud Storage for Backup Data

Introduction

Organizations generate large amounts of important data and require reliable backup storage. Cloud storage provides scalable, durable, and secure storage without requiring organizations to maintain large physical storage infrastructure.

Data Durability

Cloud storage provides high durability by protecting stored data against hardware failures. Data is stored using reliable storage systems so that the possibility of permanent data loss is very low.

Data Redundancy

Redundancy means maintaining multiple copies of data. Copies can be stored across different storage devices or locations. If one copy becomes unavailable, another copy can be used for recovery.

Encryption

Encryption converts backup data into an unreadable format. Data can be encrypted both when stored in the cloud and when transferred through a network. This prevents unauthorized users from understanding sensitive backup information.

Access Control

Access control ensures that only authorized users can access backup data. Different permissions can be provided to administrators, employees, and backup operators. Multi-factor authentication can provide additional security.

Cost Advantages

Cloud storage reduces the need to purchase physical storage devices, backup servers, and data-center infrastructure. Organizations can increase or decrease storage based on their requirements and generally pay according to their usage.

Disaster Recovery Example

If a company's data center is damaged because of a fire, flood, or hardware failure, the organization can restore its applications and databases from cloud backups. This helps the organization resume its operations quickly.

Benefits

- Reliable backup storage.
- Protection against data loss.
- Strong security.
- Easy scalability.
- Lower infrastructure cost.
- Faster disaster recovery.

3. CDN and Edge Servers for Global Applications

Introduction

A mobile application with users in different countries must provide fast and reliable service. CDN and edge servers help deliver application content closer to users and reduce network latency.

Content Delivery Network

A **Content Delivery Network (CDN)** consists of multiple servers located in different geographical regions. Frequently accessed content such as images, videos, files, and application resources is cached on these servers.

Reducing Latency

When a user requests content, the CDN delivers it from a nearby server instead of the main server. Since the data travels a shorter distance, response time is reduced and application performance improves.

Edge Servers

Edge servers are located close to end users and can deliver content or perform certain processing near the user. This reduces the need to communicate with a distant central data center.

Global Availability

Cloud providers operate data centers in multiple regions and availability zones. Applications can be deployed across different regions so that users can access services from nearby locations. If one region fails, traffic can be redirected to another region.

Real-World Example

In a real-time gaming application, players from different countries need fast responses. CDN and edge servers can deliver game content from nearby locations, reducing delay and improving the gaming experience.

Benefits

- Reduces latency.
- Improves application speed.
- Reduces server workload.
- Supports global users.
- Improves availability.
- Provides a better user experience.

4. Cloud Security and Compliance for Financial Institutions

Introduction

Financial institutions handle sensitive information such as customer details, bank accounts, payment information, and transaction records. Therefore, strong security and regulatory compliance are essential when using cloud services.

Regulatory Compliance

Cloud providers offer security controls, compliance frameworks, auditing facilities, and identity management services. These features help financial organizations meet required security and regulatory standards.

Logging

Logging records important activities such as user login, database access, file modifications, administrative actions, and financial transactions. Logs are useful for auditing and investigating security incidents.

Monitoring

Monitoring continuously checks the cloud environment for unusual activities, system failures, and security threats. Alerts can be generated when suspicious activities are detected.

Encryption

Encryption protects sensitive financial data by converting it into an unreadable format. Data should be encrypted both when stored and while being transferred between systems.

Access Control

Identity and Access Management ensures that only authorized users can access sensitive information. Role-based permissions and multi-factor authentication can provide additional protection.

Data Sovereignty

Data sovereignty requirements can be satisfied by selecting appropriate cloud regions where the organization's data is legally allowed to be stored and processed. The organization should also follow applicable data protection regulations.

Example

When a customer transfers money through online banking, transaction information is encrypted and access is restricted to authorized systems. Logging records the transaction, while monitoring tools can detect unusual transactions.

Benefits

- Protects sensitive financial data.
- Supports regulatory compliance.
- Provides auditing capabilities.
- Detects security threats.
- Controls unauthorized access.

5. Containers and Kubernetes

Introduction

Containerization allows applications to be packaged with their required libraries and dependencies. It provides a consistent environment for developing, testing, and deploying applications.

Containers vs Virtual Machines

Containers are lightweight because they share the host operating system kernel. They require fewer resources and can start quickly. Virtual machines include a complete guest operating system, so they generally require more memory, storage, and processing power.

Kubernetes Orchestration

Kubernetes is a container orchestration platform used to manage multiple containers. It automatically deploys containers, monitors their health, restarts failed containers, and manages application resources.

Scaling

Kubernetes can increase or decrease the number of containers according to application demand. During high traffic, more containers can be created. When traffic decreases, unnecessary containers can be removed.

Portability

Containers can run consistently across different environments. The same container can be deployed on different cloud providers or on-premises infrastructure with minimal changes. This improves portability and reduces dependency on a single cloud platform.

Microservices Example

An online shopping application can be divided into different microservices such as:

- User Management
- Product Catalog
- Shopping Cart
- Payment
- Order Management
- Notification

Each service can run in a separate container. Kubernetes manages these containers and scales individual services according to their workload.